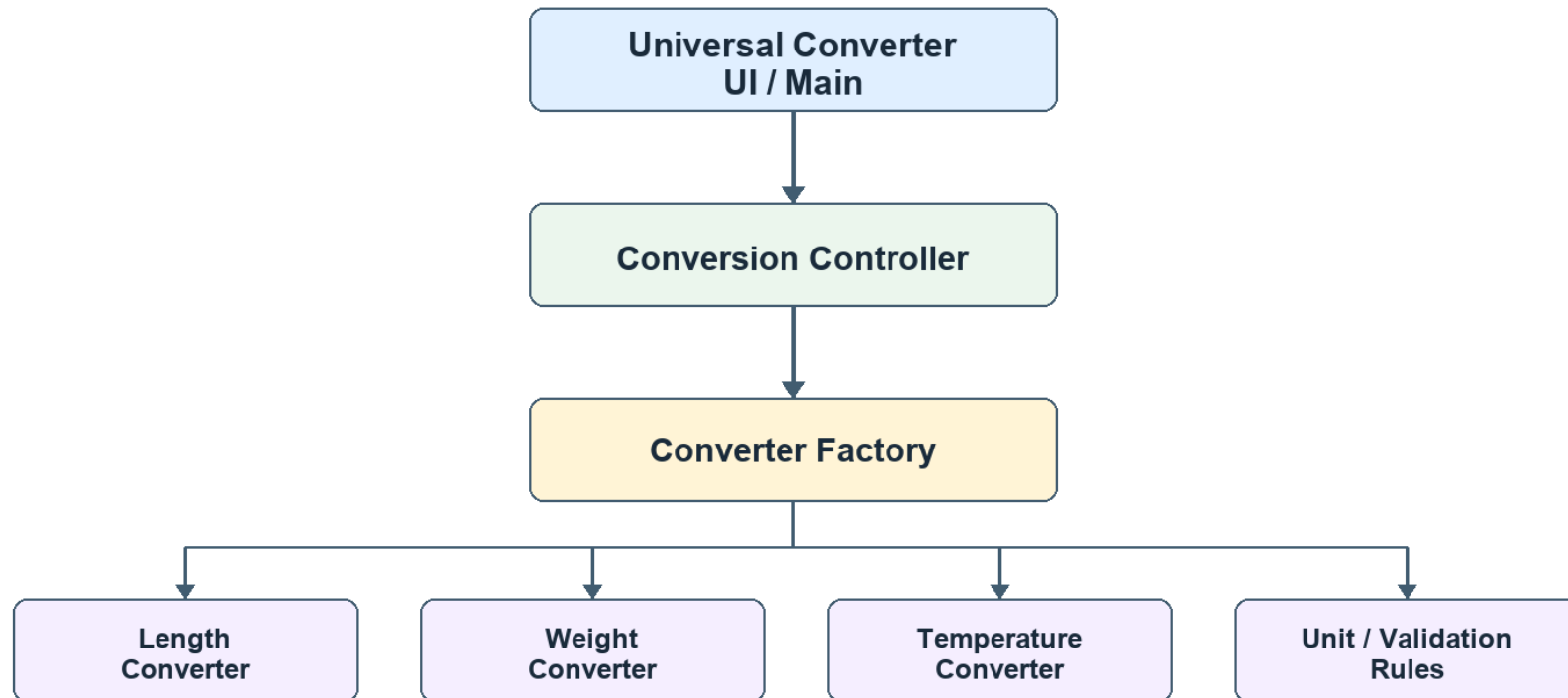


Program Structure



Top-down Integration

Unit under test	List of Stubs/Drivers	result	Stub's return value	Expected results	Actual results	Pass/Fail/No run
UI -> Controller	Stub: ConverterFactory	km to m	LengthConverter returns 1250	1.25 km -> 1250 m	1250 m	Pass
UI -> Controller	Stub: ConverterFactory	kg to g	WeightConverter returns 2000	2 kg -> 2000 g	2000 g	Pass
UI -> Controller	Stub: ConverterFactory	C to F	TempConverter returns 77	25 C -> 77 F	77 F	Pass
Controller -> Factory	Stub: LengthConverter	m to km	returns 0.75	750 m -> 0.75 km	0.75 km	Pass
Controller -> Factory	Stub: WeightConverter	g to kg	returns 0.5	500 g -> 0.5 kg	0.5 kg	Pass
Controller -> Factory	Stub: TempConverter	F to C	returns 0	32 F -> 0 C	0 C	Pass
UI -> Controller	Stub: ValidationRule	same unit	accepts unit pair	10 m -> 10 m	10 m	Pass

Unit under test	List of Stubs/Drivers	result	Stub's return value	Expected results	Actual results	Pass/Fail/No run
UI -> Controller	Stub: ValidationRule	zero input	accepts 0	0 kg -> 0 g	0 g	Pass
UI -> Controller	Stub: ValidationRule	invalid unit	rejects 'xyz'	error: unsupported unit	error shown	Pass
UI -> Controller	Stub: ValidationRule	bad category	rejects pair	error: incompatible units	error shown	Pass

Bottom-up Integration

Unit under test	List of Stubs/Drivers	result	Stub's return value	Expected results	Actual results	Pass/Fail/No run
LengthConverter	Driver: ConverterFactory	km to m	driver calls convert(1.25)	1250 m	1250 m	Pass
LengthConverter	Driver: ConverterFactory	m to km	driver calls convert(750)	0.75 km	0.75 km	Pass
WeightConverter	Driver: ConverterFactory	kg to g	driver calls convert(2)	2000 g	2000 g	Pass
WeightConverter	Driver: ConverterFactory	g to kg	driver calls convert(500)	0.5 kg	0.5 kg	Pass
TemperatureConverter	Driver: Controller	C to F	driver calls convert(25)	77 F	77 F	Pass
TemperatureConverter	Driver: Controller	F to C	driver calls convert(32)	0 C	0 C	Pass
ValidationRule	Driver: Controller	same unit	driver calls validate(m,m)	valid	valid	Pass
ValidationRule	Driver: Controller	zero input	driver calls validate(0)	valid	valid	Pass
ValidationRule	Driver: Controller	invalid unit	driver calls validate(xyz)	unsupported error	error returned	Pass

Unit under test	List of Stubs/Drivers	result	Stub's return value	Expected results	Actual results	Pass/Fail/No run
Factory + Converters	Driver: UI workflow	full length flow	select length, km, m	3.5 km -> 3500 m	3500 m	Pass